

Augment, then split: a leakage-free replication of a multi-task attention model for mammography

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Independent research project

Technical report, September 24, 2026 · Code and results:

<https://github.com/sweetmAGIcian7/mammography-multitask-ai>

Abstract

Esen et al. (IEEE Access, 2025) propose a multi-task convolutional network with channel–spatial attention (CBAM) that classifies mammograms as benign or malignant and grades breast density at the same time. They report 93.6% accuracy and an AUC of 0.962. I re-implemented the model and audited the evaluation. The paper’s images were augmented into ~ 72 rotated and flipped copies *before* the train/test split, and the reported accuracies are arithmetically impossible on a test set of original images. Re-running that protocol on the same public data, the replication scores a perfect AUC of 1.000; with a patient-level split it scores 0.755, and density prediction falls to the majority-class baseline. On CBIS-DDSM (2,802 mammograms, 1,460 patients, biopsy-confirmed labels) with patient-level cross-validation, the paper’s design reaches AUC 0.782 (95% CI 0.762–0.802), 0.746 on the official test split, and density agreement QWK 0.769. Joint training gives no measurable benefit over single-task models, and CBAM adds a small +0.013 AUC. Density grading transfers to an external digital dataset (INbreast, QWK 0.660), while calibration does not. Scored against 376 radiologist-drawn lesion outlines, the CBAM “attention map” localises lesions barely better than a lesion-blind brightness map (pointing game 0.24 vs 0.18). Grad-CAM of the same model does localise them (0.44). None of the paper’s attention statistics could be reproduced. The architecture is reasonable; the published numbers mostly measure the evaluation protocol.

Same architecture, different evaluation: the published number measures leakage

EfficientNet + CBAM, two heads (malignancy, density). Hollow dot: value reported in the paper (no CI given).

● Leaky: copies of test images in training ● Partly leaky: same patient in training ● Leakage-free: patient-level split

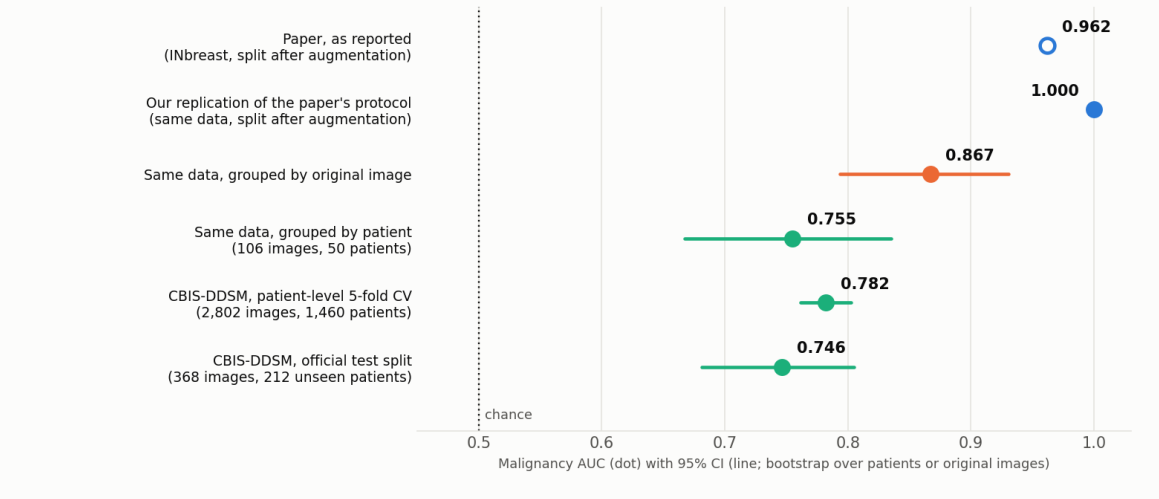


Figure 1: Malignancy AUC of the same architecture under each evaluation protocol. Blue: the test set contains augmented copies of training images. Orange: grouped by image, but other views of the same patient remain in training. Green: no patient in both training and test. Lines are 95% bootstrap CIs.

1 Introduction

Mammography screening generates large volumes of images, and deep-learning models that flag suspicious findings or estimate breast density are an active research area. Breast density matters twice: dense tissue both hides tumours and is an independent risk factor. A model that predicts malignancy *and* density from one image is therefore attractive, and multi-task learning [2] suggests that the two tasks might help each other.

Esen et al. [1] build such a model: an EfficientNet backbone [3], a CBAM attention module [4] and two task heads. On INbreast [5] they report 93.6% malignancy accuracy (AUC 0.962) and 88.9% density accuracy, together with attention maps that “strongly concentrate on irregular mass margins”. My first re-implementation reached only 34–53% accuracy. Instead of tuning until the numbers matched, I examined how the paper evaluated its model and designed experiments to test each claim separately:

Q1 Can the evaluation protocol alone explain the reported numbers? (Sec. 4.1)

Q2 With a leakage-free protocol, how good is the design, and do multi-task learning and attention help? (Sec. 4.2)

Q3 Are the probabilities trustworthy, and does the model transfer to a different dataset? (Sec. 4.3, 4.4)

Q4 Do the attention maps actually point at lesions? (Sec. 4.5)

The critique is meant as a good-faith replication. Augment-then-split and patient overlap are common in medical imaging [8, 9], and nothing here implies intent.

2 Audit of the reference evaluation

The full, itemised audit is in `PAPER_AUDIT.md` in the repository. Its main points:

- **Augmentation before splitting.** Sec. III-B states that every image was rotated to 11 angles and flipped, and that “subsequent to the augmentation process” the data were split (Table 1: 7,208 augmented images). Fig. 1 cites the Huang & Lin dataset [7]: 106 INbreast mass images augmented to 7,632 files in exactly the eight density×pathology folders of the paper’s Table 1. The training set therefore almost certainly contains rotated copies of every test mammogram.
- **The accuracies need an augmented test set.** A 20% hold-out of 410 images has 81–83 images. With that many images the achievable accuracies near the reported values are 0.926–0.928 and 0.938–0.940. The reported 0.936 (hold-out) and 0.931/0.933/0.934 (cross-validation folds) cannot occur, so they must have been measured on thousands of augmented files.
- **Image-level split.** INbreast has 115 patients and 410 images; stratifying by image places the other views of the same breast on both sides of the split.
- **Brier score and ECE contradict each other.** For any binned calibration estimate, Jensen’s inequality gives $\text{Brier} = \mathbb{E}[(p - y)^2] \geq \sum_b w_b (\bar{p}_b - \bar{y}_b)^2 \geq (\sum_b w_b |\bar{p}_b - \bar{y}_b|)^2 = \text{ECE}^2$. The reported pair (Brier 0.010, ECE 0.365–0.419) violates this bound by more than an order of magnitude.
- **Other inconsistencies.** Table 1 rows sum to 7,328, not the stated 7,208. The ROC figure and the results table give different AUCs for the same models. Two different attention architectures and three augmentation recipes are described. Two of the four “correct attention” examples in Fig. 4 are misclassified.

3 Methods

Data. Three public datasets were used (Table 1). The Huang & Lin set reproduces the paper’s setting for Q1. CBIS-DDSM [6] gives biopsy-confirmed pathology for every finding, a BI-RADS density grade and patient IDs. INbreast serves only as an external test set. CBIS-DDSM stores a mammogram twice when it contains both a mass and calcifications. These 55 duplicates were merged (malignant if any finding is malignant). Patients listed in both official CBIS-DDSM splits (22) were treated as training patients.

Table 1: Datasets. “Patients” is the unit of every split and every bootstrap.

Dataset	Images / patients	Labels	Role
Huang & Lin [7]	106 originals × 72 copies / 50	benign/malignant, density	Q1: leakage experiment
CBIS-DDSM [6]	2,802 / 1,460 (official test 368 / 212)	biopsy pathology, density	Q2–Q4: training, CV, test
INbreast [5]	410 / 108	density, BI-RADS (proxy)	Q3: external test only

Preprocessing. Each mammogram is decoded at reduced resolution. The breast is cropped with an Otsu threshold and the largest connected component. The image is flipped so the chest wall is on the left, then resized with its aspect ratio kept to 640×384 pixels and zero-padded. INbreast DICOMs are first mapped to 8 bits with a percentile window. The transform is recorded, so lesion masks can be sent through exactly the same geometry (Sec. 4.5).

Model. Fig. 2 shows the replicated architecture (paper Sec. IV-B). The loss is $2.0 \cdot \text{BCE}_{\text{malignancy}} + 0.8 \cdot \text{CE}_{\text{density}}$ with label smoothing 0.05, as in the paper. I used EfficientNet-B0 rather than B3 and 640×384 rather than 224×224 input. Mammographic findings are small, and B3 at full resolution was beyond the free GPU budget. Four variants share

Replicated model (paper Sec. IV-B): one network, two outputs

Ablations switch off CBAM (mt_plain) or one head (st_path, st_dens); everything else identical.

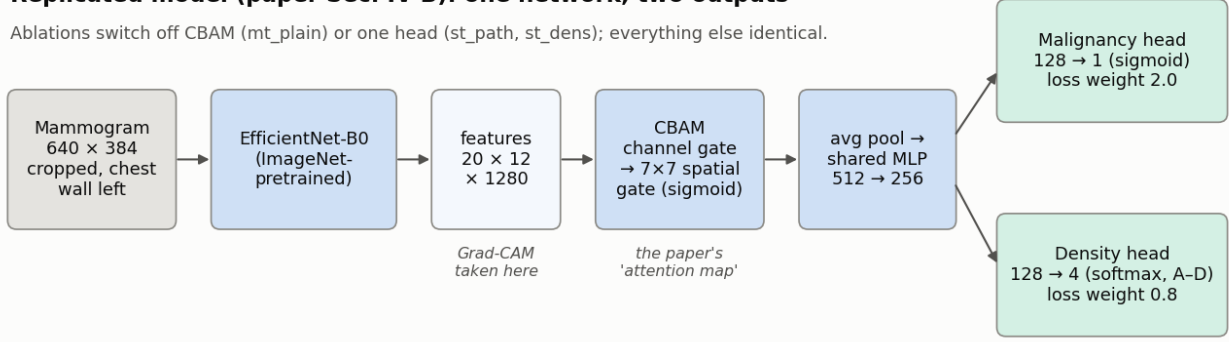


Figure 2: The replicated model. The CBAM spatial gate is what the paper calls the attention map. Grad-CAM is computed on the backbone output, which is the same layer for every variant.

everything else: mt_cbam (paper design), mt_plain (no CBAM), st_path (malignancy head only) and st_dens (density head only). Training: AdamW ($\text{lr } 2 \cdot 10^{-4}$, one-cycle schedule), 15 epochs, batch 16, mixed precision, and augmentation that respects the chest-wall orientation (vertical flip, $\pm 10^\circ$ rotation, zoom, shift, brightness/contrast). There is no early stopping and no tuning on test folds.

Evaluation. All splits are by patient and asserted to be disjoint. CBIS-DDSM uses 5-fold patient-level cross-validation with identical folds for all variants; metrics are pooled over out-of-fold predictions. 95% confidence intervals come from a bootstrap over patients (2,000 resamples). Variants are compared with a *paired* bootstrap on the same test images. Density is scored with the quadratic-weighted kappa (QWK), which respects the order $A < B < C < D$. Every metric is reported next to a majority-class baseline.

4 Results

4.1 Q1: the leakage experiment

The paper-design model was trained three times on the Huang & Lin files. Only the split changed: (i) random over files, as in the paper; (ii) grouped by original image; (iii) grouped by patient (5-fold CV, 8 epochs).

Table 2: Q1. Same model, same data, only the split changes. Mean $\pm$ SD over 5 folds; pooled AUC with a cluster-bootstrap CI.

Split	Test patient also in training	Malignancy AUC	Pooled AUC (95% CI)	Density acc.
Paper protocol (augment, then split)	100%	1.000 ± 0.000	1.000	1.000
Grouped by original image	97%	0.867 ± 0.066	0.867 (0.794 to 0.931)	0.689
Grouped by patient	0%	0.724 ± 0.093	0.755 (0.667 to 0.835)	0.403
Majority class		0.500		0.401

The paper’s protocol yields a perfect score, higher than the paper’s own 0.962, because on average 57 of each test file’s 71 siblings are in training. Grouping by image is not enough: 97% of test images still had a different view of the same patient in training. With a patient-level split, malignancy AUC falls to 0.72–0.76, and density accuracy (0.403) equals the majority-class rate (0.401). With leakage, the model had learned to recognise *patients*, not density.

4.2 Q2: a leakage-free multi-task model on CBIS-DDSM

Table 3: Q2. CBIS-DDSM, patient-level 5-fold CV, pooled out-of-fold (2,802 images, 1,460 patients). 95% CIs by patient bootstrap.

Variant	Malignancy AUC	Malignancy acc.	Density acc.	Density QWK
mt_cbam (paper design)	0.782 (0.762 to 0.802)	0.700	0.668	0.769 (0.746 to 0.791)
mt_plain (no attention)	0.769 (0.748 to 0.789)	0.697	0.653	0.761 (0.735 to 0.784)
st_path (malignancy only)	0.773 (0.752 to 0.793)	0.691	–	–
st_dens (density only)	–	–	0.681	0.780 (0.755 to 0.802)
Majority class	0.500	0.552	0.395	0.000

With an honest split the design reaches AUC 0.782 (Table 3), within the range usually seen for whole-image classification of downsampled mammograms. On the official test split (368 images, 212 unseen patients), a single

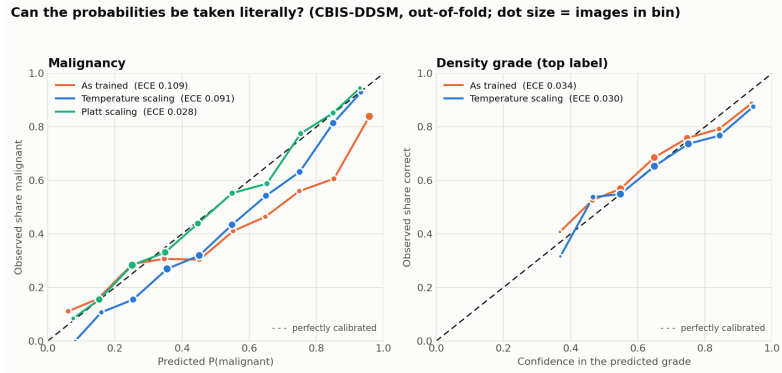


Figure 3: Q3a. Reliability of the malignancy output and of the density grade (cross-validation, out-of-fold), before and after cross-fitted temperature and Platt scaling.

model trained on the official training patients reaches AUC 0.746 (0.681 to 0.805) and QWK 0.711 (0.631 to 0.773). Paired comparisons:

- adding the density task to malignancy: $\Delta\text{AUC} +0.009$ (-0.004 to $+0.024$), no clear effect;
- adding the malignancy task to density: $\Delta\text{QWK} -0.010$ (-0.026 to $+0.005$), no clear effect;
- adding CBAM for malignancy: $\Delta\text{AUC} +0.013$ ($+0.002$ to $+0.026$). This is a small gain from a single training seed, below the fold-to-fold spread (± 0.025);
- adding CBAM for density: $\Delta\text{QWK} +0.008$ (-0.005 to $+0.023$).

Performance depends on the lesion type: AUC 0.83 on images with masses and 0.72 on images with only calcifications, which are a few pixels across at this resolution. CC and MLO views score the same (0.78).

4.3 Q3a: calibration and operating points

A model can rank cases well and still give misleading probabilities (Fig. 3). Using the cross-validation out-of-fold predictions of Table 3, every calibration map was fitted on four folds and applied to the fifth.

- The raw malignancy output is over-confident *and* biased: its mean prediction is 0.542 against a true rate of 0.448, and ECE is 0.109 (95% CI 0.095–0.131).
- Temperature scaling [10] ($T \approx 1.74$) only softens the probabilities (ECE 0.091). Platt scaling [11], which also has an intercept, removes the bias: ECE 0.028, Brier 0.205 $\rightarrow$ 0.188.
- The density head is already calibrated (top-label ECE 0.034).
- The parameters transfer only partly to a separately trained model: on the official-split model, ECE drops from 0.206 to 0.090.
- A threshold chosen on the other folds for 90% sensitivity reached 89.8% on unseen patients, with specificity 0.386.

The sensitivity/specificity trade-off is set by this threshold. The paper claims that its loss weights “ensure minimizing false negatives”, but the relative weight of two task losses does not set that trade-off. All our models satisfy $\text{Brier} \geq \text{ECE}^2$.

4.4 Q3b: external validation on INbreast

The four official-split models were applied unchanged to all 410 INbreast images, which are full-field digital mammograms from Portugal (Fig. 4). The training data are scanned US film from the 1990s.

- **Density transfers.** QWK is 0.660 (0.550 to 0.743) against 0.711 in-domain. 93.4% of predictions are within one grade of the radiologist’s. The majority-class accuracy is 0.357; the model reaches 0.538. It over-calls the densest grade (61 predicted D vs 28 true).
- **Malignancy, on a proxy.** INbreast has no biopsy label for most images, so the target is BI-RADS 4–6 vs 1–3. The AUC is 0.822 (0.761 to 0.877), and the mean score rises steadily with BI-RADS: 0.25, 0.28, 0.28, 0.43, 0.75, 0.79. This task is easier than CBIS-DDSM’s, because it includes normal mammograms.
- **Calibration does not transfer.** The two multi-task variants rank images similarly, but on normal INbreast images they predict 0.25 and 0.54 on average. The CBIS temperature makes density ECE slightly worse (0.107 $\rightarrow$ 0.129).
- Multi-task vs single-task density: $\Delta\text{QWK} +0.059$ (-0.019 to $+0.146$). This suggests but does not show better transfer.

Trained on scanned film (CBIS-DDSM), tested on digital mammograms (INbreast)

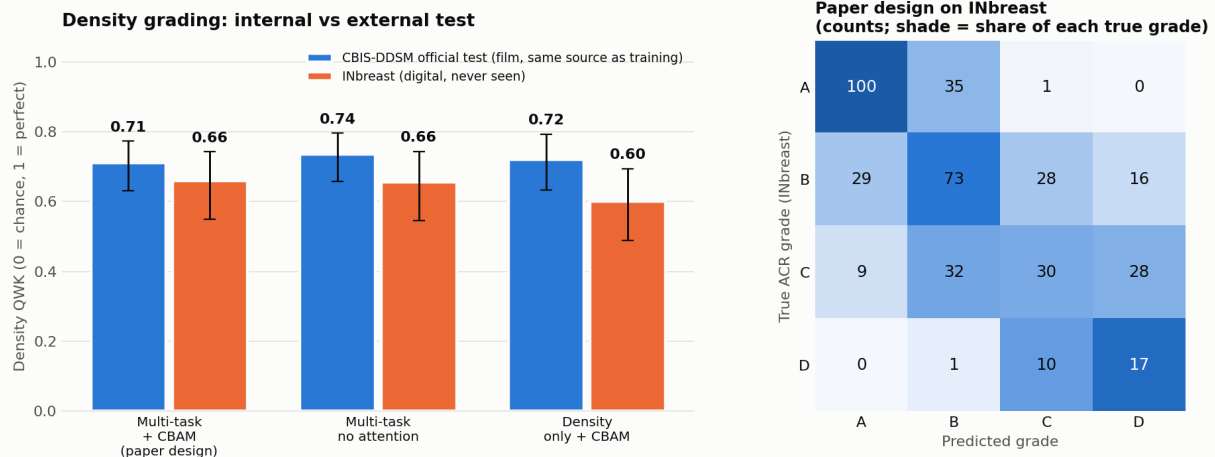


Figure 4: Q3b. Density grading on the internal (CBIS-DDSM official test) and external (INbreast) test sets for each variant, with 95% CIs (bootstrap over patients), and the paper design’s confusion matrix on INbreast.

4.5 Q4: do the attention maps point at the lesion?

The paper supports its interpretability claims with four hand-picked images and statistics without a stated method. I scored the CBAM gate and Grad-CAM [12] of the official-split models on the official CBIS-DDSM test split against the radiologists’ ROI masks. Masks were identified by content, since CBIS-DDSM sometimes swaps its crop and mask files, and sent through each image’s own crop, flip and resize: 376 of 390 findings, on 354 images from 206 patients. Maps are 20×12 (stride 32). Metrics are computed inside the breast:

- the pointing game [13]: the map’s maximum lies on the lesion, within one cell;
- energy in lesion: the share of the map’s mass on the lesion;
- pixel AUC: lesion vs other breast pixels.

Lesion-blind baselines (uniform, random, brightest tissue, breast centre) and an ImageNet-only network define what “better than chance” means.

Table 4: Q4. Localisation on 354 official-test images (95% CI, patient bootstrap). Chance energy = the lesion’s share of the breast (0.018).

Map	Pointing (± 1 cell)	Energy in lesion	Pixel AUC
CBAM gate, paper design	0.24 (0.19 to 0.29)	0.019 (0.016 to 0.024)	0.69 (0.66 to 0.72)
Grad-CAM (malignancy), paper design	0.44 (0.38 to 0.50)	0.103 (0.085 to 0.121)	0.72 (0.68 to 0.76)
Grad-CAM (malignancy), no-attention model	0.42 (0.36 to 0.48)	0.106	0.71
Grad-CAM (density), paper design	0.19	0.031	0.53
<i>ImageNet-only network, Grad-CAM</i>	0.15	0.028	0.61
<i>Brightest tissue</i>	0.18 (0.14 to 0.22)	0.022	0.67
<i>Uniform (chance)</i>	0.08	0.018	0.50
<i>Ceiling: ROI mask at 20×12</i>	1.00	0.431	1.00

The CBAM gate is *open almost everywhere*: its mean value is 0.85, and its normalised entropy is 0.995. It beats the brightness baseline by only +0.055 (0.000 to +0.108) on pointing, and its energy on the lesion equals chance. A min–max-scaled heatmap of such a gate can still *look* focused. Grad-CAM of the same model points at the lesion in 44% of images (66% of malignant ones) and puts $5.7\times$ chance energy on it. The ImageNet-only control scores 0.15, so the localisation comes from training on mammograms. CBAM does not improve Grad-CAM localisation (+0.023, -0.017 to +0.061). Lesions smaller than one cell are found far less often (0.29 vs 0.56 for 1–4 cells).

Using explicit definitions on the map pooled to the paper’s 7×7 grid, none of the paper’s statistics was reproduced:

- Gini, malignant vs benign: 0.088 vs 0.081 (paper 0.68 vs 0.43);
- entropy: 3.876 vs 3.879 nats, both 99.6% of the maximum $\ln 49$ (paper 2.31 vs 3.74; the paper’s “benign” value is itself 96% of the maximum);
- peak attention, dense vs non-dense: 0.999 vs 0.999, $r = +0.02$ (paper 0.82 vs 0.61, $r = 0.61$);
- 80%-mass area: 75% vs 75% (paper 12% vs 29%).

Do the maps point at the lesion? CBIS-DDSM official test, 354 images with ROI masks
shaded rows: maps that know nothing about lesions, and the best a 20x12 map could do

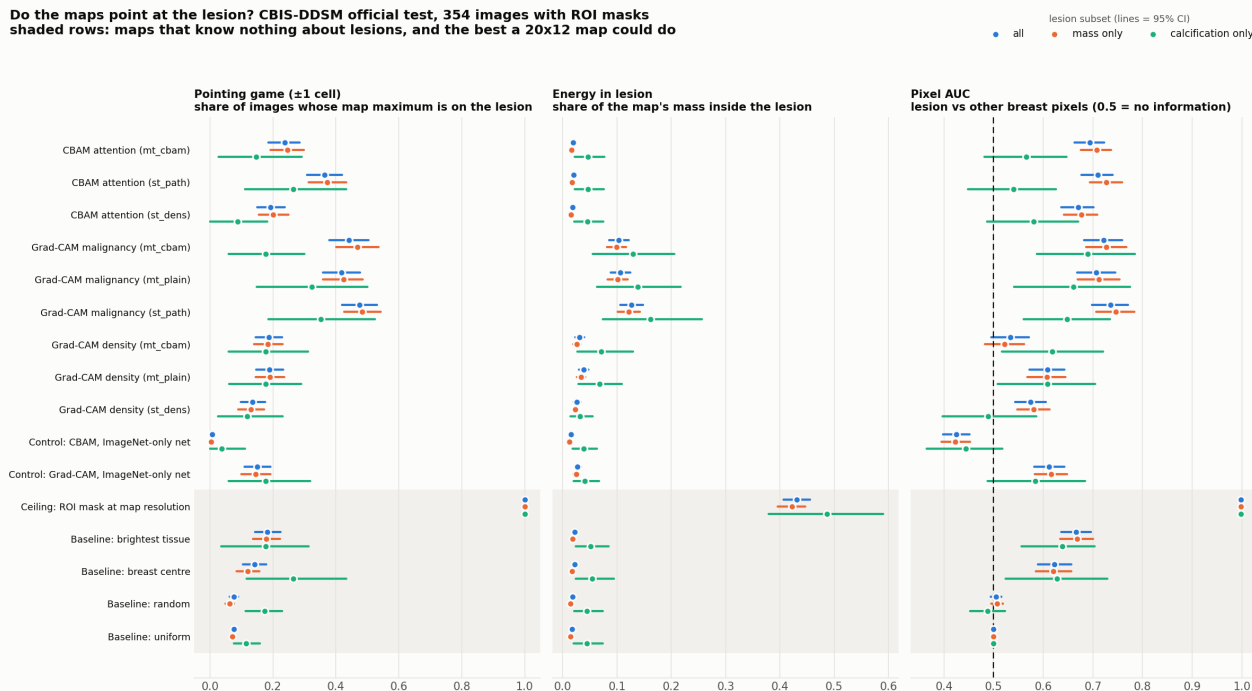


Figure 5: Q4. Pointing game (± 1 cell) and energy in lesion for each map, with 95% CIs (patient bootstrap), next to lesion-blind baselines and the ceiling (the ROI mask itself at map resolution).

5 Discussion

What the design is worth. With an honest evaluation the multi-task CBAM network is a solid baseline: AUC 0.75–0.78 for whole-image malignancy on CBIS-DDSM, good density agreement, and density grading that survives a change of country and imaging technology. Its selling points are not supported. Multi-task training does not improve either task measurably. It saves one network, which is a real but modest benefit. The attention gate adds little accuracy and no interpretability. Where explanations are needed, Grad-CAM on the same model is the better tool, although it too misses more than half the lesions at this resolution.

Why the gap matters. The difference between 0.96 and 0.75–0.78 is not a detail. A reader of the paper would expect a model close to clinical performance and a working lesion localiser; neither holds. The cause is a single ordering decision, augmenting before splitting, combined with an image-level split on a dataset of 50 patients. Both are cheap to avoid: split by patient first, augment only the training set, and report a majority baseline and a confidence interval with every metric.

Limitations.

- **One seed per model.** Effects of about 0.01 AUC (CBAM) are not established.
- **Not the paper’s exact setup.** The backbone and resolution differ, and the paper’s data and splits are unreleased, so Q1 relies on the dataset the paper cites.
- **Findings, not screening.** CBIS-DDSM contains only images with findings, so these results say nothing about screening performance.
- **Small and uneven test data.** The external set is small (108 patients) and uses a BI-RADS proxy for malignancy. Calcifications are under-represented in the localisation test (34 images), and their outlines are loose.

6 Conclusion

Reproducing a published medical-imaging result required auditing its evaluation first. The reported 93.6% accuracy is explained by the paper’s protocol: the same model on the same data scores 100% under that protocol and about 75% without leakage. Evaluated properly, the architecture is useful but ordinary, and its attention maps do not show what the paper says they show. All code, per-image predictions, figures and an online demo (<https://mammography-multitask-ai.streamlit.app>) are public.

Reproducibility and AI assistance. Every experiment is one script and one free Kaggle notebook. Retraining with the same seed reproduced the official-split results to three decimals. Unit tests check the split logic, preprocessing geometry

and metrics. The code and much of the analysis were produced with an AI assistant (Claude, Anthropic) under the author’s direction; the author ran all experiments and reviewed every result.

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